



AI Idea Forge

Week 3 – System Integration & Workflow Automation

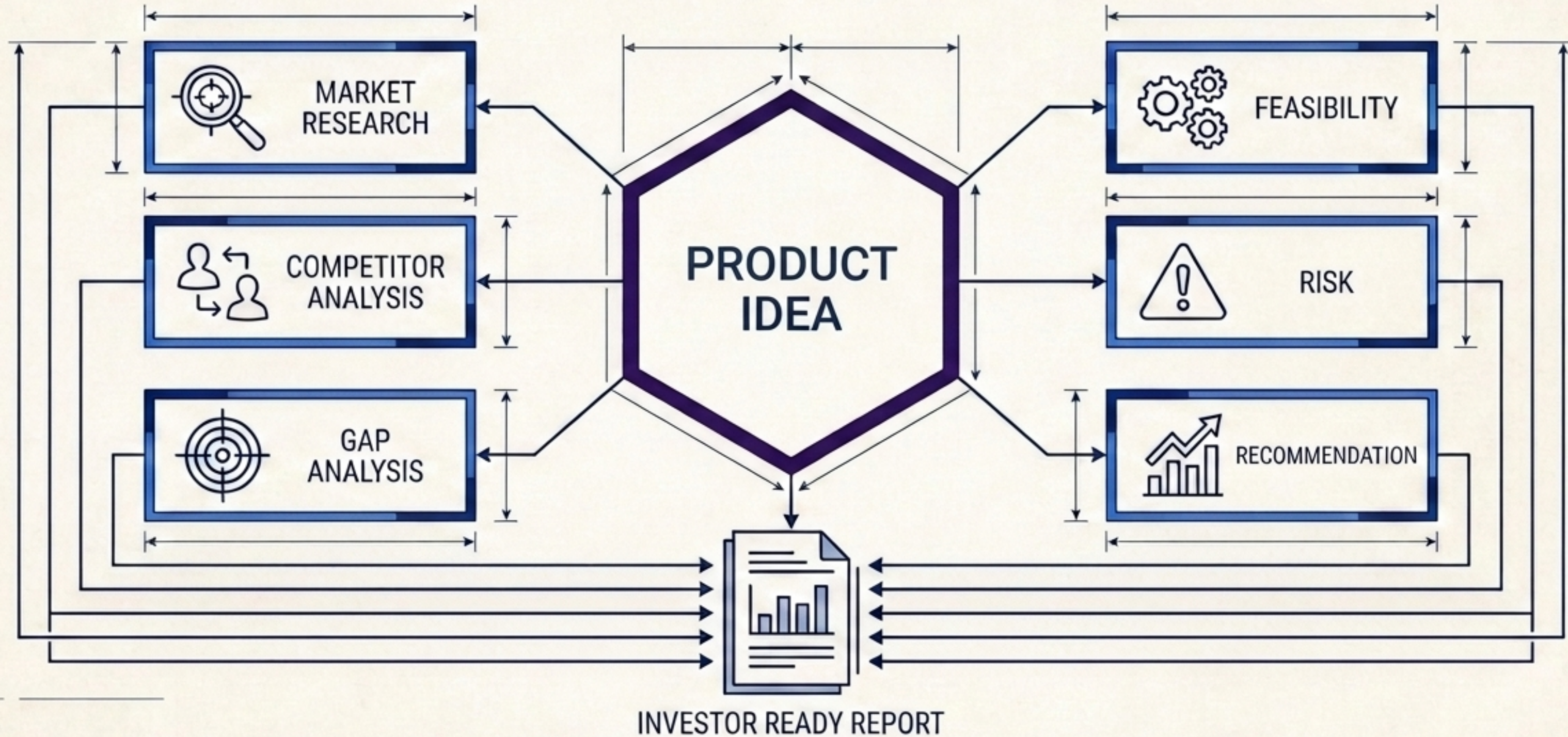
Batch 4 – Group 8
August 2026

Transforming raw ideas into validated, investor-ready products through AI orchestration.

AI Idea Forge: Project Overview

An AI-powered Startup Validation Platform automating entrepreneurial research.

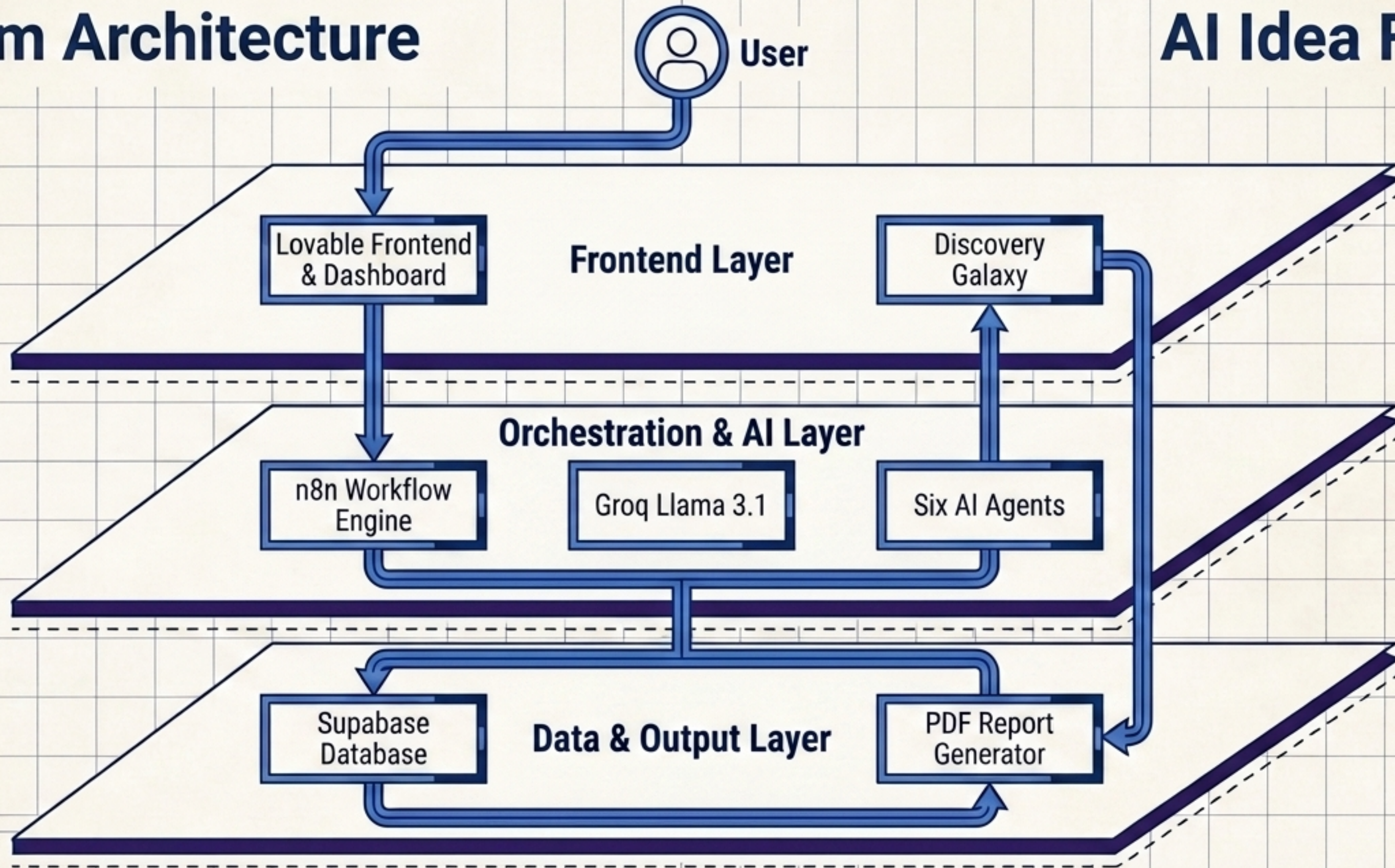
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Multiple specialized AI agents work in parallel to evaluate a single startup idea from every strategic angle.

System Architecture

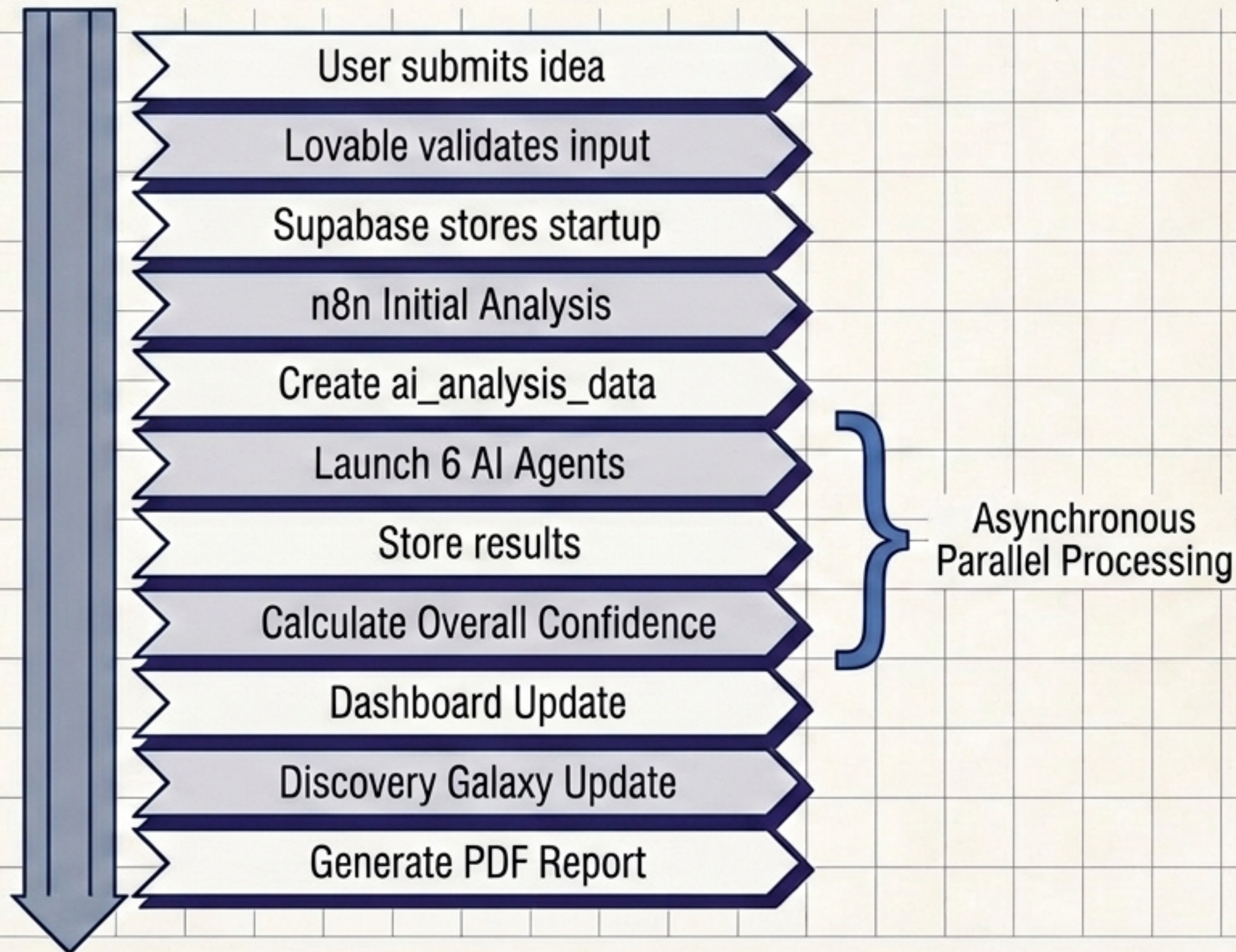
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Strict separation of concerns between frontend UI, workflow orchestration, AI inference, and database persistence.

End-to-End Workflow Pipeline

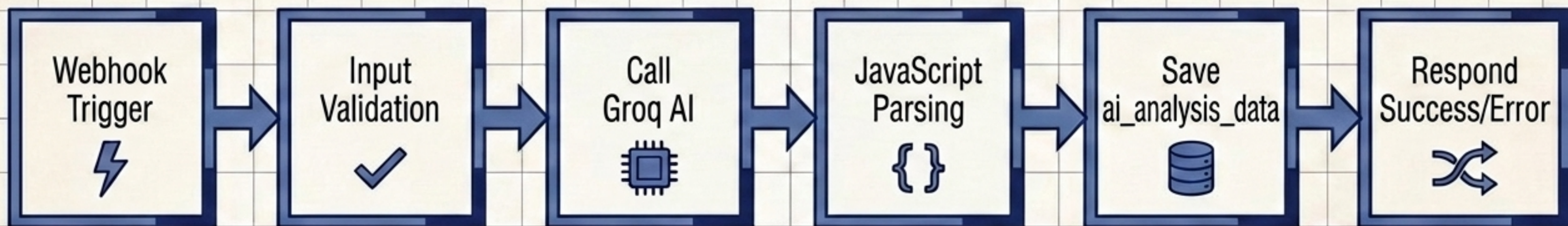
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A fully automated pipeline drives the user's idea from initial submission to final interactive visualization without manual intervention.

Initial Analysis Workflow

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This workflow establishes the master analysis record.

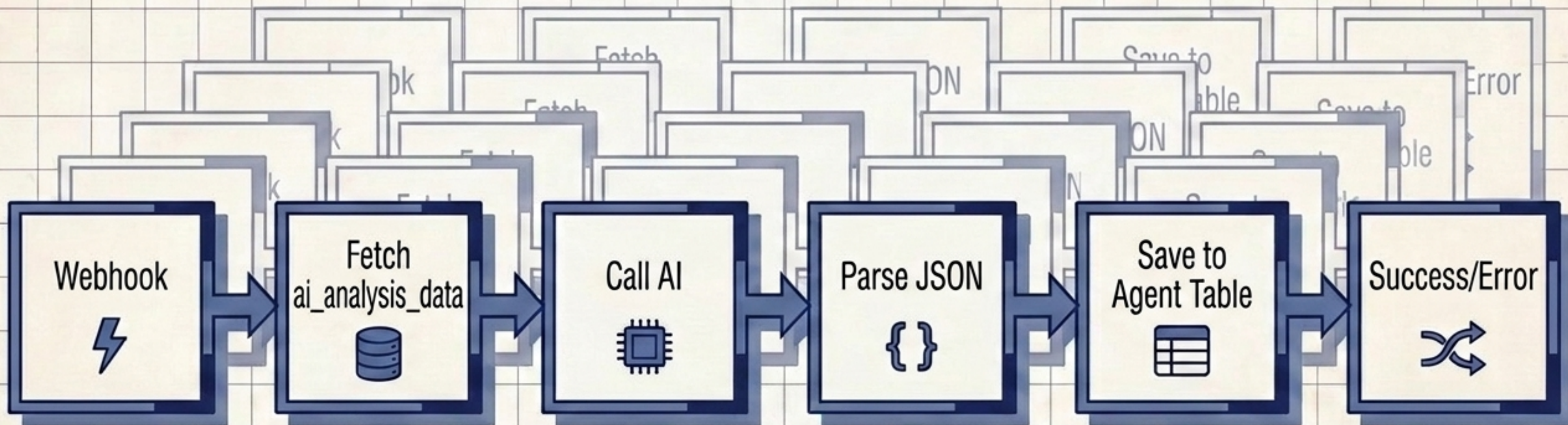


Key Takeaway: This 'ignition switch' workflow standardizes raw user input into a master analysis record before specialized agents take over.

Generic AI Agent Workflow

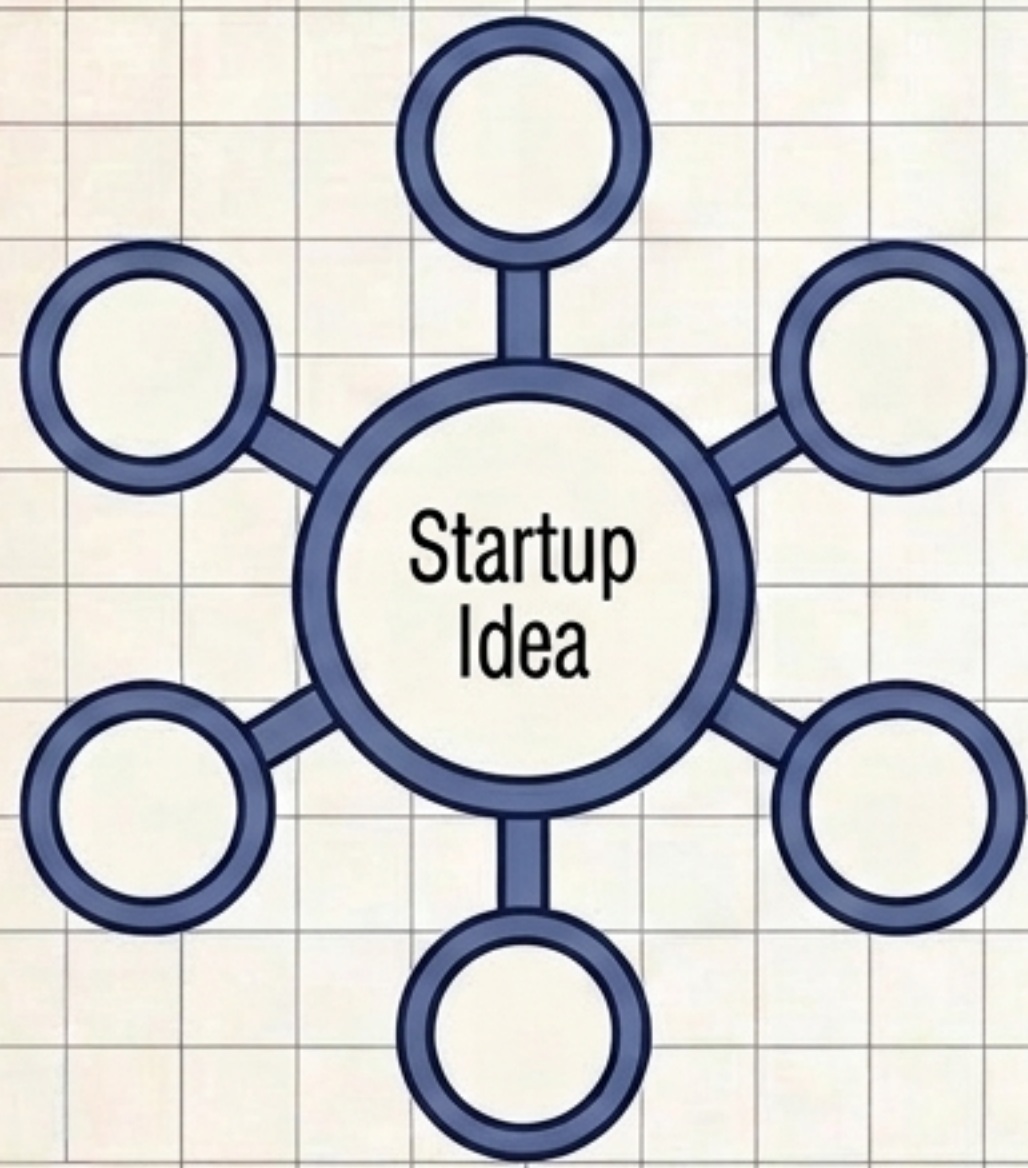
A single, robust n8n template cloned for all six specialist agents.

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Key Takeaway: Standardizing the agent workflow ensures massive scalability, allowing new AI agents to be added with zero architectural changes.

AI Agent Orchestration AI Idea Forge

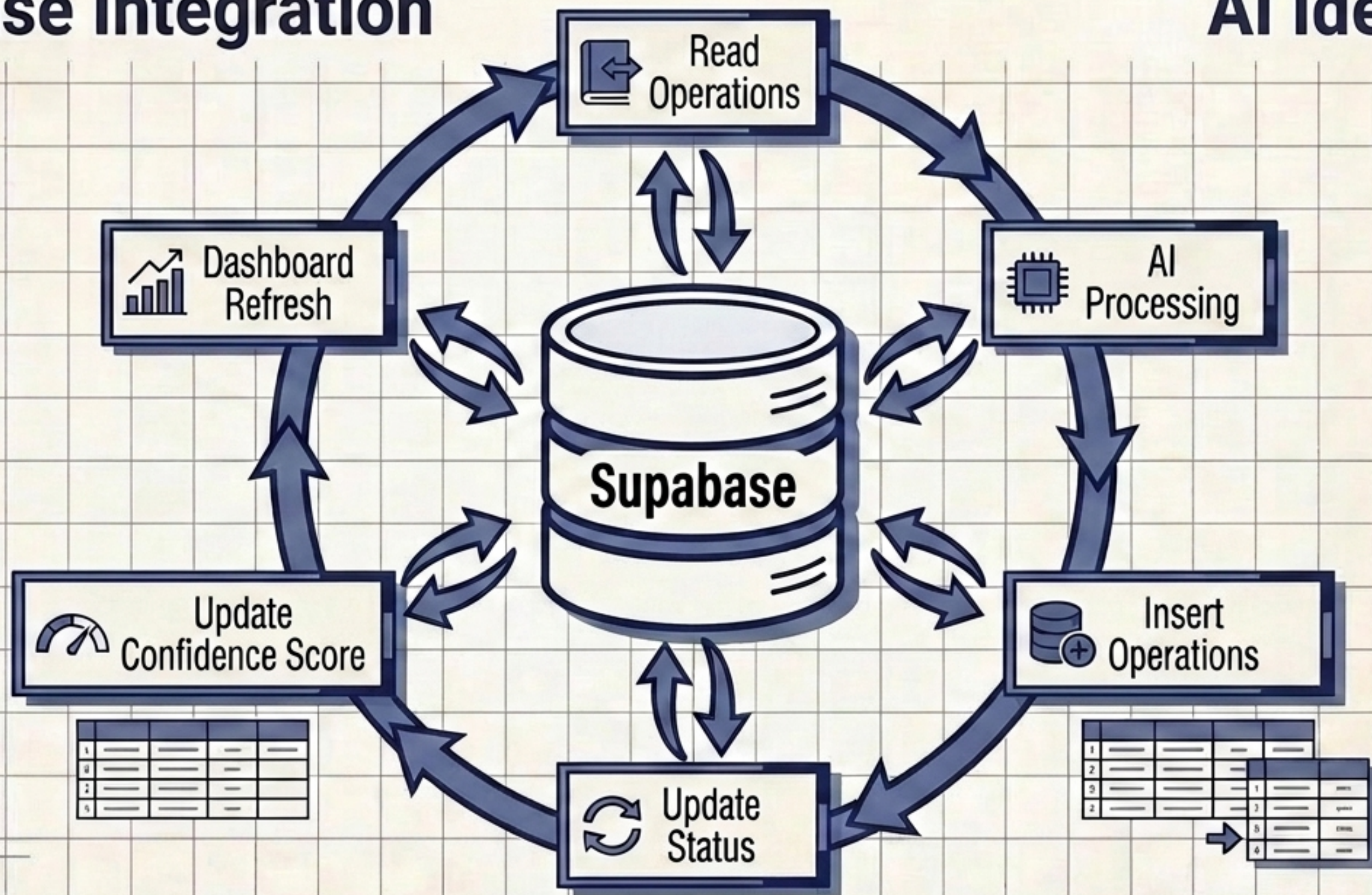


AI Agent	Responsibility
Market Research	Evaluates market demand, customer segments, trends, and market size
Competitor Analysis	Identifies competitors and market positioning
Gap Analysis	Identifies unmet customer needs and innovation opportunities
Feasibility Analysis	Evaluates technical and business feasibility
Risk & Challenge	Identifies business and technical risks
Recommendation Engine	Generates strategic recommendations

Key Takeaway: Dividing analysis into strict business domains dramatically improves LLM output quality and system maintainability.

Database Integration

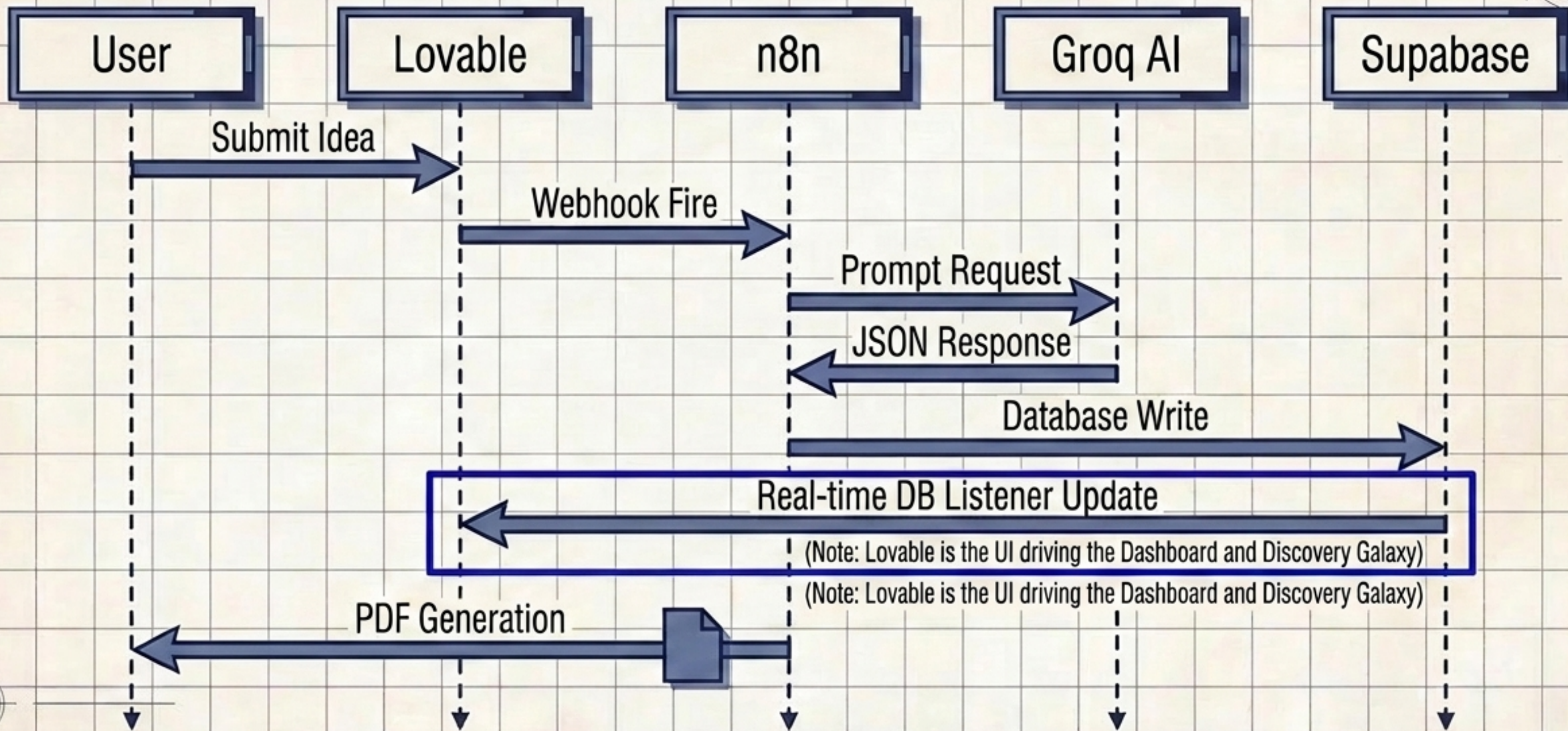
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More info on Supabase - refer "AI-Idea-Forge-Supabase-Database-Design" document

Lovable ↔ n8n Integration

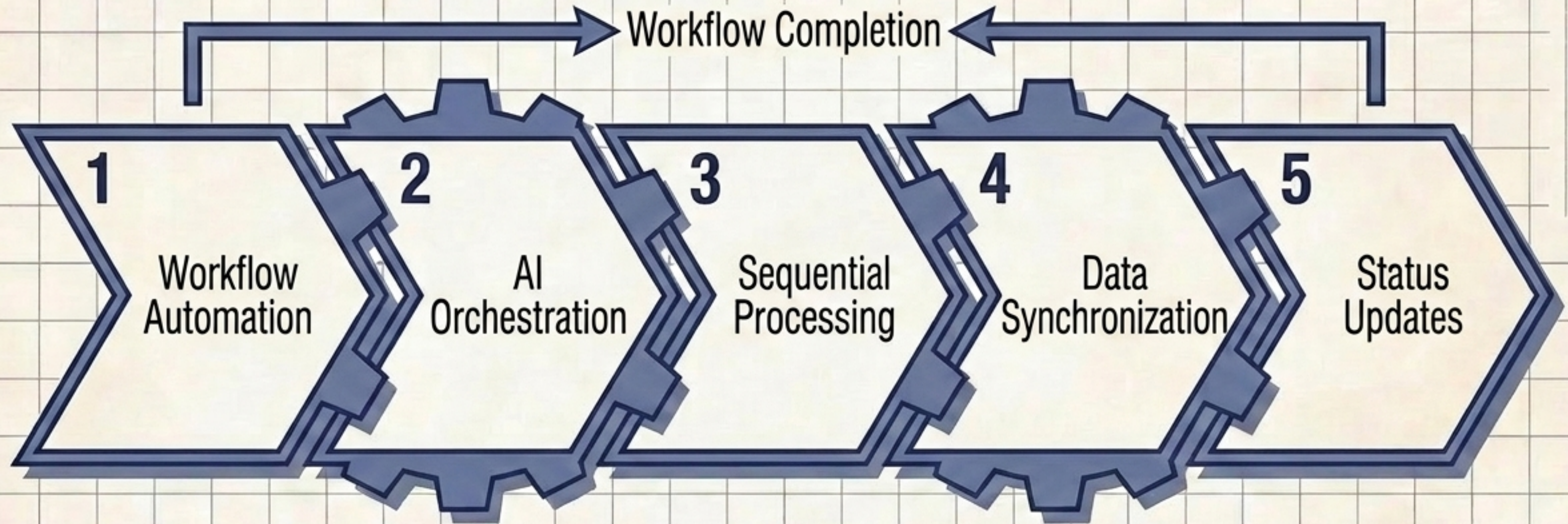
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Key Takeaway: Asynchronous webhook triggers and real-time database listeners decouple the UI from long-running AI inference times.

Workflow Automation

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Key Takeaway: n8n coordinates the entire pipeline, transforming isolated AI calls into a cohesive, synchronized assembly line.

Discovery Galaxy

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Live AI Agent Status

Progress Tracking

Overall Confidence Score

85%

Figure – Discovery Galaxy Multi-Agent Analysis Interface

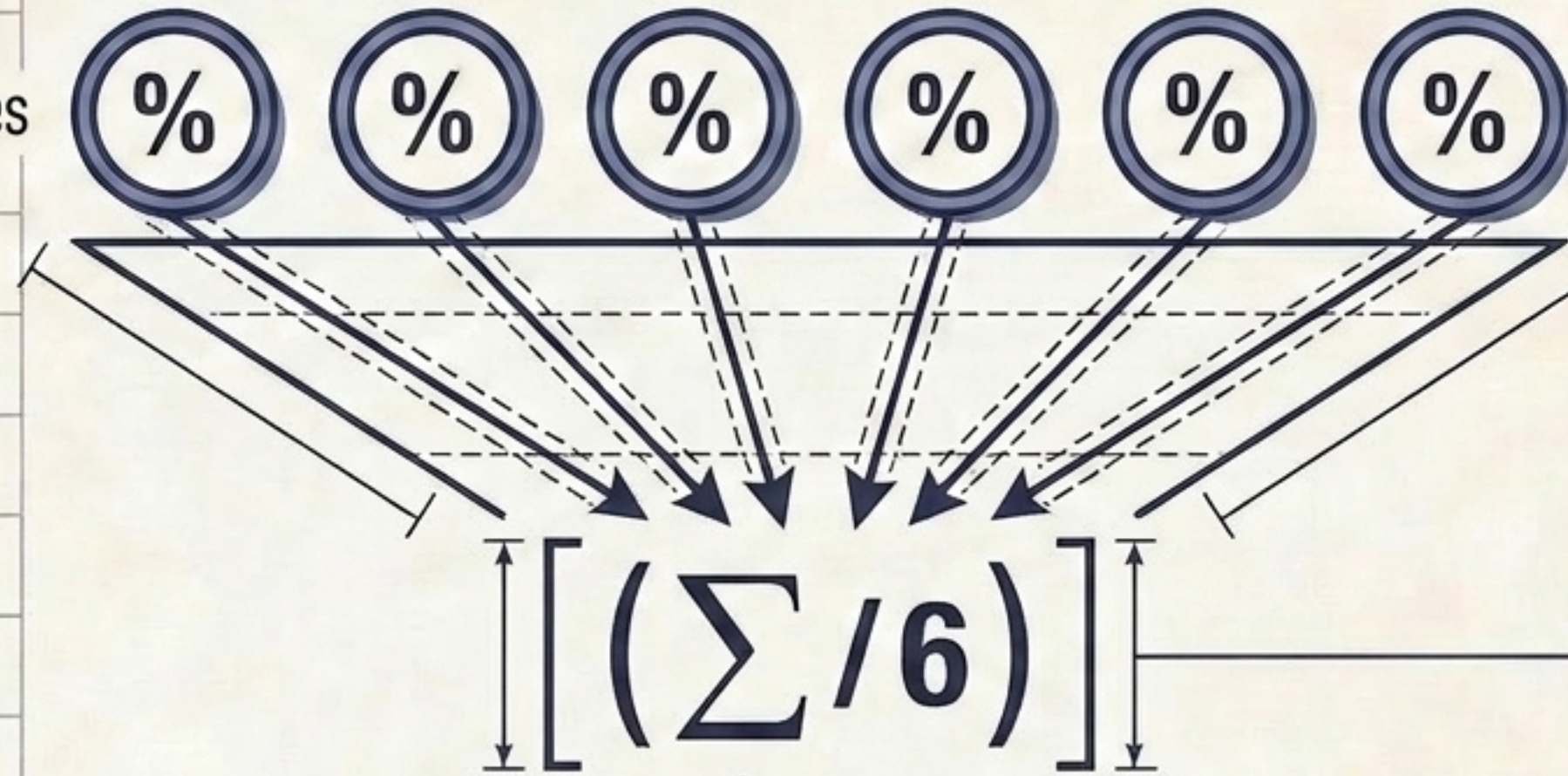
Key Takeaway: The complex asynchronous backend is visualized as a transparent, watchable process, building user trust in the AI's reasoning.

Confidence Score Calculation

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6 AI Agent Scores

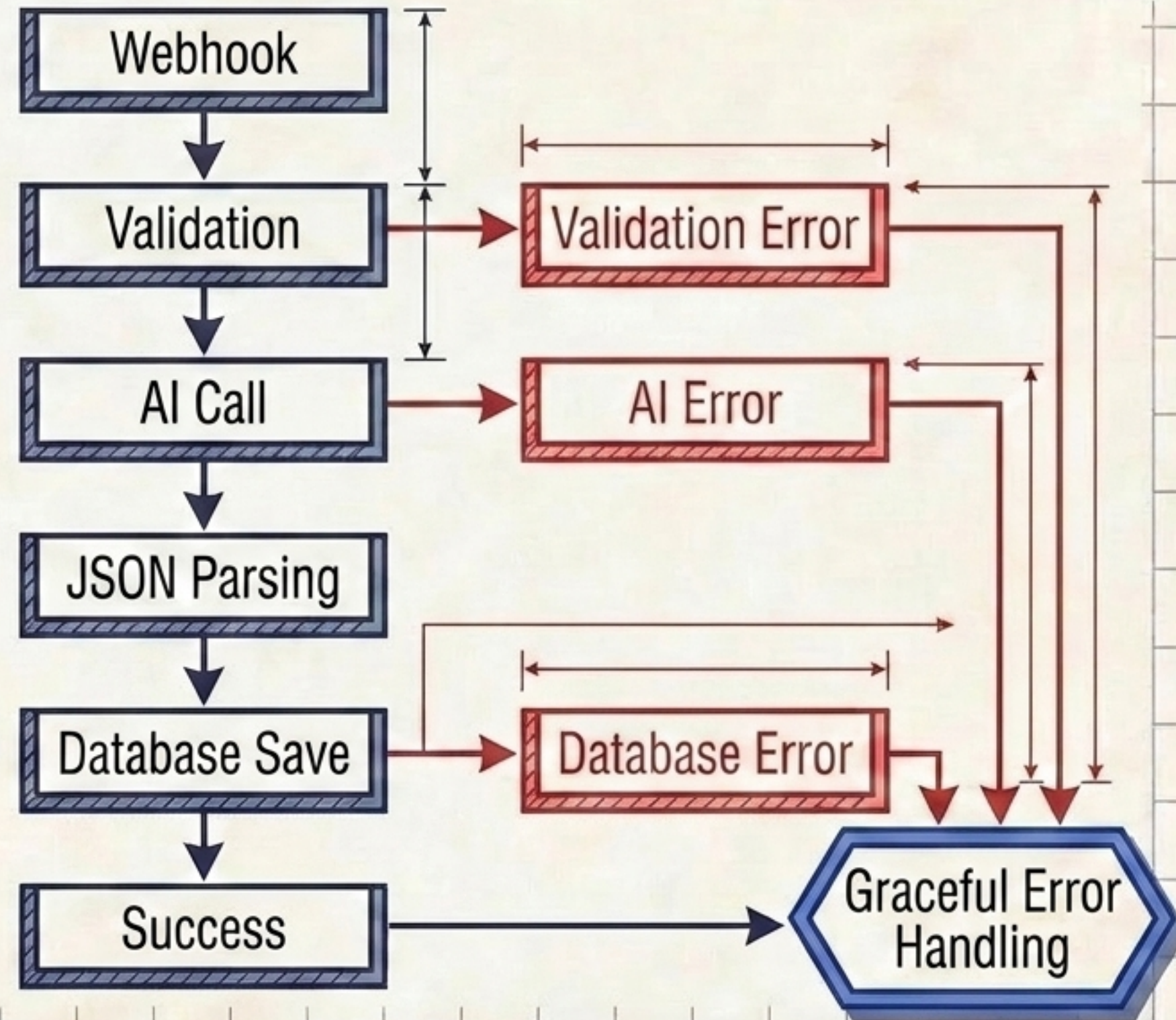


Overall Confidence Score

Key Takeaway: The overall confidence score is a dynamic arithmetic mean of all six completed, independent AI agent evaluations.

Error Handling

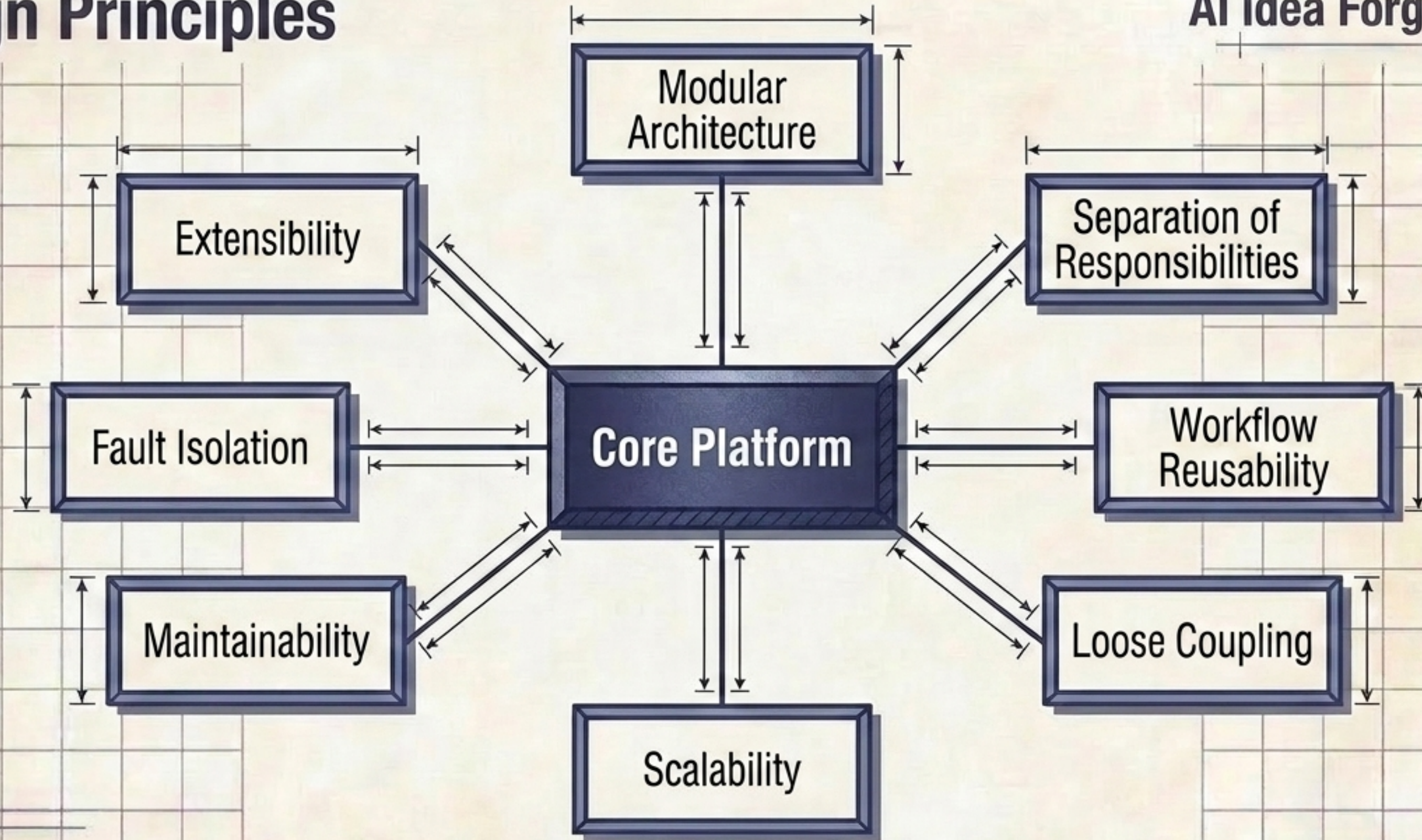
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Anticipating failure states—from malformed LLM JSON to webhook retries—ensures robust, user-friendly system stability.

Design Principles

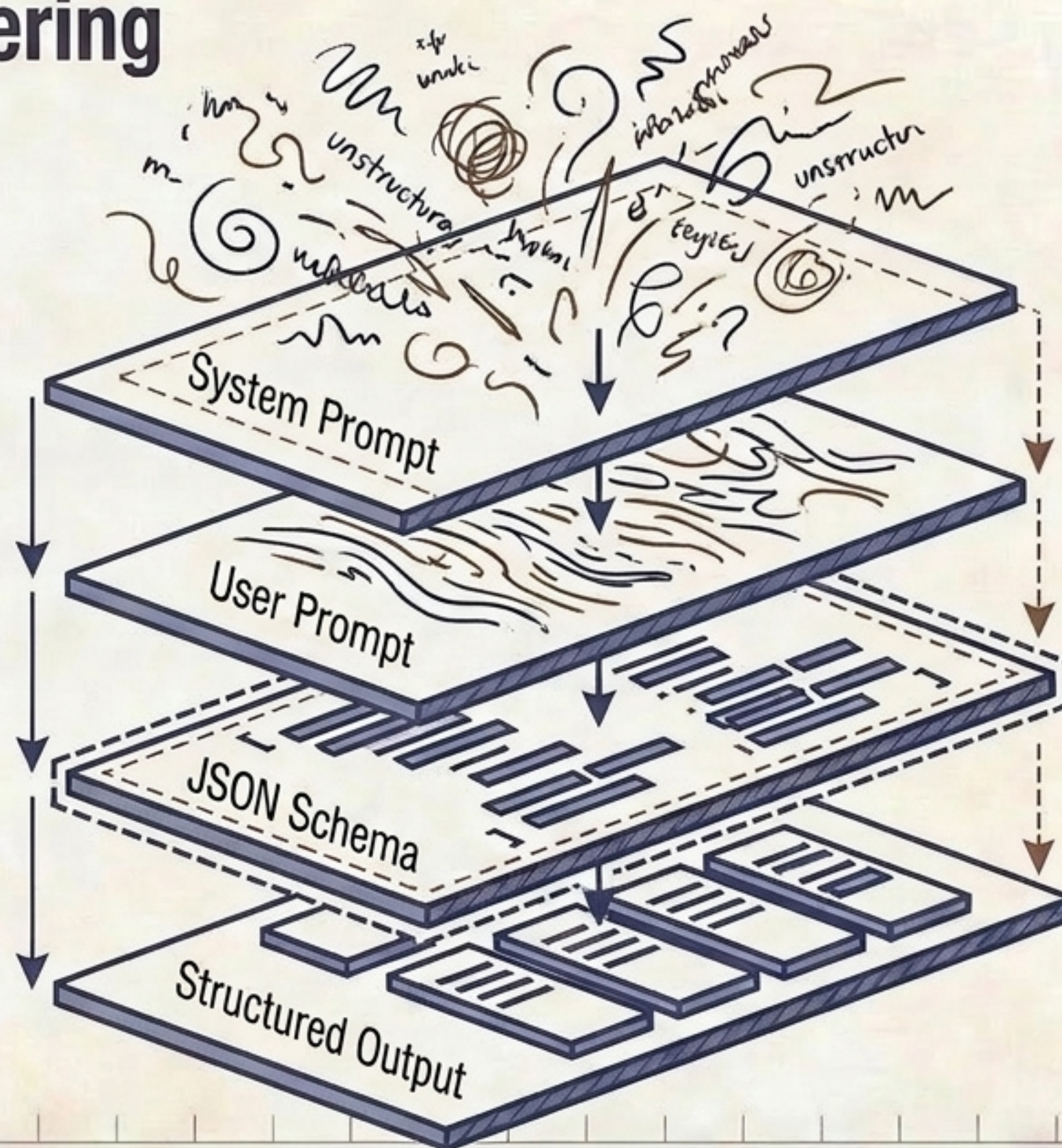
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Key Takeaway: Every architectural decision prioritizes long-term system scalability and isolation of volatile AI components.

AI Prompt Engineering

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Key Takeaway: Constraining generative AI with strict JSON schemas is the bridge between unpredictable language models and reliable relational databases.